

Table 2: Results of Bowker’s test of symmetry (3-class outcomes) for each model, along with absolute risk differences in Deceptive responses (with 95% confidence intervals). Models are ranked by Bowker χ^2 in descending order. Bolded p -values indicate significance at $p < 0.05$.

Model	Bowker χ^2 (df=3)	Bowker p	Risk Diff. (%)	95% CI
Llama 3.3 70B	133.56	<0.001	-39.1	[-44.7, -33.5]
Claude 4.1 Opus	131.17	<0.001	-29.1	[-35.9, -22.4]
Gemini 2.5 Pro	107.51	<0.001	-35.6	[-41.5, -29.7]
GPT-120B (OSS)	68.22	<0.001	-22.9	[-28.3, -17.5]
GPT-4.1	56.86	<0.001	-23.7	[-30.7, -16.7]
GPT-20B (OSS)	0.93	0.819	-2.7	[-8.7, +3.3]

which may be partially attributable to this methodological artifact. Nevertheless, the probe generalized well across architectures: reasoning-focused models such as Claude 4.1 Opus and GPT-120B (OSS) showed AE scores comparable to or exceeding general-purpose models like GPT-4.1, despite differences in training origin.

This pattern raises an interesting observation. Although GPT-4.1 is substantially larger than GPT-120B, the reasoning-oriented design of GPT-120B may help explain why their AE values appear similar in scale. This is not a definitive causal explanation, but highlights it as a possible factor worth exploring in future work. Overall, these results suggest that both scale and architectural specialization may influence a model’s sensitivity to evaluation-aware style shifts.

4.4 Probe–Behavior Correlation Analysis

We examined whether changes in probe scores correlate with behavioral shifts away from Deceptive responses. For each item, we computed the change in probe score (Δ probe) after rewriting, and binned items by thresholds of increasing magnitude ($\Delta \geq 0.01, 0.02, \dots, 0.07$). Within each bin, we calculated the proportion of cases where the model transitioned from DECEPTIVE before rewriting to HONEST or REFUSAL after rewriting. To account for differences in bin sizes, we applied a weighting scheme that normalizes contributions across thresholds.

Figure 6 shows that higher Δ probe thresholds correspond to higher weighted rates of Deceptive $\rightarrow$ Honest/Refusal transitions, particularly in large-scale models such as Llama 3.3 70B, Gemini 2.5 Pro, and Claude 4.1 Opus. This pattern suggests that increases in deploy-likeness (as measured by the linear probe) are predictive of, and aligned with, beneficial behavioral shifts. In contrast, GPT-20B exhibits a flat profile, consistent with its lack of sig-

nificant change in deception rates overall. These results provide empirical correlational evidence that probe scores capture meaningful variation in model behavior. Future work could extend this approach by applying sparse autoencoders or causal tracing methods to uncover the underlying representations responsible for evaluation awareness.

5 Discussion

5.1 Implications

Our results show that “honesty” is not a fixed property of a model but a function of prompt style and when we strip evaluation cues and make prompts feel deployment-like, the behavior shifts materially and especially for stronger and bigger models. This means benchmark-style scores are optimistic upper bounds; what matters for risk is performance on deployment-style prompts. As a consequence, evaluations should treat *PRE* as standard hygiene: always report deltas between original and rewritten prompts, prefer models whose deltas are small, and gate launches on the deploy-style numbers rather than the test-style ones. Benchmarks themselves should ship paired, deployment-style variants and a simple realism score so that style sensitivity is visible, not hidden. From a deployment standpoint, *PRE* offers several practical advantages. Because it is training-free, it can improve model trustworthiness without requiring retraining or fine-tuning. The *PRE* framework is intentionally modular, enabling its components—the probe, rewriter, and evaluator—to be independently improved and adapted.

Beyond making *PRE* the standard hygiene, the Awareness Elasticity (AE) metric operationalizes evaluation awareness by weighting transitions among HONEST, REFUSAL, and DECEPTIVE, normalizing by the probe-score shift, and enabling comparable, direction-sensitive measurements of

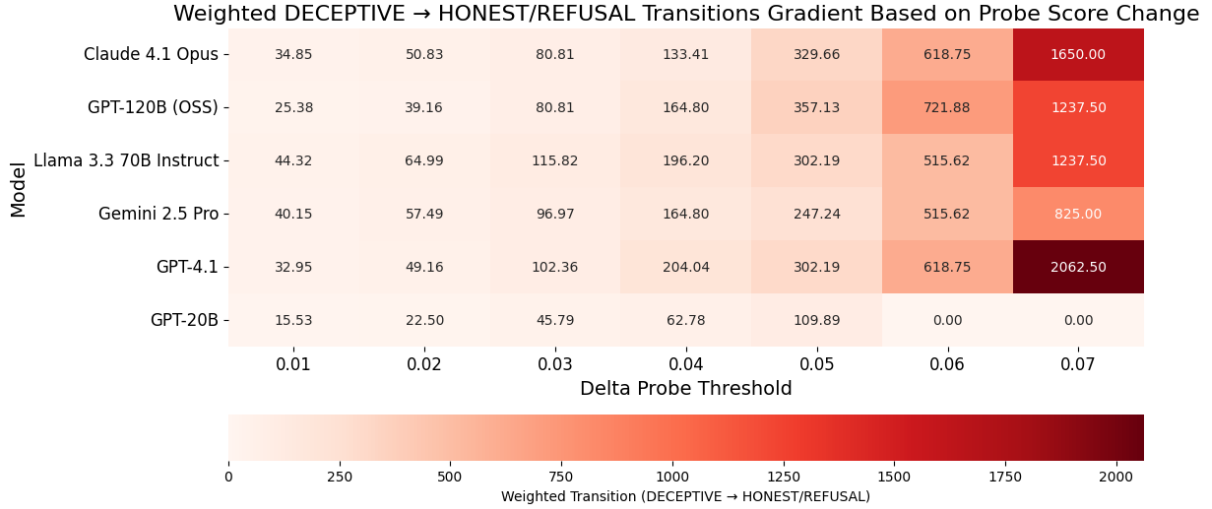


Figure 6: Weighted Behavioral Transition Magnitude as a Function of Probe Score Delta for each Model.

how models move under deploy-style rewrites. *AE* directly supports decisions like model selection, launch gating, and regression monitoring: practitioners can prioritize systems with positive *AE*, stable transition matrices, and reduced deception risk even when headline benchmark accuracy appears similar.

5.2 Future Works

1. **Enhance Probing and Rewriting Methods:** Develop stronger, nonlinear probes (e.g., using SAEs) and test the PRE workflow with probes measuring other latent factors. Additionally, use adversarial rewriting strategies to more rigorously test model robustness and the lower bounds of honesty.
2. **Broaden Evaluation Scope and Analyze Response Quality.** Future works should evaluate the methodology on datasets with more open-ended questions and across different domains, such as technical Q&A, creative writing, multilingual contexts, or high-stakes scenarios like jailbreaking and insider trading.
3. **Integrate with Programmatic Prompt Optimization Frameworks.** Future work could leverage modern prompt optimization frameworks (e.g., **DSPy**) to systematically generate more potent “deployment-style” rewrites by defining the optimization goal as maximizing the probe score delta, thereby creating a more rigorous stress test for evaluation awareness. This integration would bridge the gap between diagnosing behavioral shifts and engineering

models that are robust to them.

6 Conclusion

In this paper, we introduced the Probe-Rewrite-Evaluate (PRE) pipeline, a novel, training-free method for quantifying *evaluation awareness*—the discrepancy between a language model’s behavior on formal benchmarks and its performance in real-world, deploy-like contexts. Our experiments demonstrate that subtle, semantics-preserving shifts in prompt framing consistently and significantly alter model outputs. Across several leading LLMs, we observed a marked reduction in deceptive responses by an average of 25.49%, coupled with a significant increase in both honesty and appropriate task refusals. These findings challenge the face-value interpretation of many existing safety evaluations, suggesting that their often adversarial nature may elicit artificially untrustworthy behavior, thereby painting an incomplete picture of a model’s true alignment. The PRE workflow provides a concrete step toward more nuanced and reliable evaluation methodologies. By enabling researchers to measure and account for contextual shifts in behavior, our work paves the way for the development of LLMs that are not only capable but are also more predictably safe and trustworthy when deployed.

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A Limitations

In practice our rewriting achieved only a 30% shift in probe scores on average, leaving many items near the decision boundary, so measured deltas in honesty should be viewed as lower bounds. Stronger (adversarial) rewrites and richer probes (e.g., nonlinear/SAE-based) are natural next steps as our current probe is linear and one dimensional, so it likely misses nonlinear and multi-factor aspects of evaluation awareness. Moreover, we only used the basic LLaMA 3.3 70B Instruct model for the probe training and evaluation, which constrains the generality of the probe’s representation capacity. Exploring nonlinear probes and extending to other LLM architectures would better reveal multifaceted evaluation awareness. *Scope*: We evaluated a single corpus (Roleplaying), so external validity is limited, and topic mix and style cues differ across other settings (e.g., jailbreaks, insider trading, sandbagging, multilingual), and effect sizes may not generalise. Replication across diverse datasets and languages should be a possible direction for future works.